

Cyber Security Fundamentals & Networking Basics

SJM Institute of Technology
Department of Computer Science &
Engineering
Student Documentation

13 Days Cyber Security Workshop • SJMIT College

1. Introduction to Cyber Security

Cyber Security is the practice of protecting systems, networks, programs, devices, and data from digital attacks, unauthorized access, damage, or disruption.

Three Pillars of Information Security (CIA Triad)

- **Confidentiality (Privacy):** Ensuring information is accessible only to authorized personnel.
- **Integrity:** Protecting data accuracy, completeness, and preventing unauthorized modification.
- **Availability:** Ensuring timely and reliable access to systems and information for authorized users.

Security Scope & Core Threat Landscape

- **Physical Security:** CCTV monitoring, biometric access control, perimeter security, and disaster recovery.
- **Software & System Security:** Firewalls, antivirus solutions, malware protection, and vulnerability patching.
- **Common Cyber Threats:** Viruses, Worms, Trojans, Malware, Ransomware, Phishing attacks, and Social Engineering.

2. Operating Systems & Server Architectures

Operating systems serve as the interface between the computer hardware and the user.

OS Category	Description & Architecture	Common Examples
End-User / Client OS	Optimized for individual user productivity, GUI navigation, and daily interactive tasks.	Windows 11, Windows 10, Windows 8/8.1, Windows 7, macOS, Desktop Linux (Ubuntu, Mint).
Server OS	Engineered for multi-user services, continuous uptime, enterprise networking, and robust security policies.	Windows Server (2025, 2022, 2019, 2016, 2012), Red Hat Enterprise Linux (RHEL), Ubuntu Server.

Common Types of Servers

#	Server Type	Function & Role
1	Web Server	Serves static content (HTML, CSS, JS, media) via HTTP/HTTPS (e.g., Apache, Nginx).
2	Web Application Server	Executes dynamic backend business logic and application code.
3	Database Server	Stores, retrieves, and manages structured and unstructured databases (e.g., MySQL, PostgreSQL).
4	Email Server	Processes, routes, and stores incoming/outgoing emails (SMTP, IMAP, POP3).

5	Gaming Server	Maintains real-time game multiplayer state and player synchronization.
6	Load Balancer Server	Distributes incoming client traffic across multiple backend server nodes.
7	Proxy Server	Acts as an intermediary forwarding outbound client requests to the Internet.
8	Reverse Proxy Server	Sits in front of web servers to provide SSL termination, caching, and attack filtering.
9	Network Server	Manages core network operations, IP routing, DHCP leases, and DNS lookups.
10	File Server	Provides centralized file storage and access control across a local area network.

3. 14 Key Domains of Cyber Security

#	Domain	Description
1	Physical Security	Securing data centers, server rooms, biometric access, and surveillance.
2	Perimeter Security	Firewalls, border gateways, and perimeter defense systems.
3	Application Security	Securing code, input validation, and protecting against OWASP Top 10 flaws.
4	IAM Security	Identity and Access Management, Role-Based Access Control (RBAC), and MFA.
5	Cloud Security	Securing cloud infrastructure, workloads, and shared responsibility models.
6	Data Security	Encryption at rest/in transit, data loss prevention (DLP), and tokenization.
7	Email Security	Anti-phishing filters, SPF, DKIM, and DMARC verification.
8	AI Security	Securing machine learning models against adversarial inputs and data poisoning.
9	OS Security	Operating system hardening, patch management, and privilege separation.
10	Hardware Security	Secure boot, TPM chips, and Hardware Security Modules (HSM).
11	Firmware Security	UEFI/BIOS protection and verifying firmware cryptographic signatures.
12	Embedded Security	Protecting microcontrollers, IoT devices, and embedded systems.
13	Container Security	Hardening Docker containers, image registries, and Kubernetes clusters.
14	Network Security	IDS/IPS, network segmentation, VPNs, and packet filtering.

Specialized Information Security Domains

- **GRC (Governance, Risk, and Compliance):** Aligning organizational security with industry standards (ISO 27001, NIST, SOC 2).
- **OT - ICS Security (Operational Technology - Industrial Control Systems):** Securing mission-critical industrial manufacturing and control systems (SCADA, PLC, RTU).

4. Ethical Hacking Teams & Prerequisites

● Red Team (Offensive Security)

Focuses on penetration testing, adversary simulation, finding vulnerabilities, and exploiting security weaknesses.

● Blue Team (Defensive Security)

Focuses on infrastructure defense, SOC monitoring, incident response, digital forensics, and proactive system hardening.

● Purple Team (Collaborative Synergy)

Facilitates continuous collaboration and knowledge sharing between Red and Blue teams to maximize organizational defense effectiveness.

Prerequisites for a Cyber Security Engineer

1. Operating Systems proficiency (Linux & Windows internals).
2. Networking fundamentals & protocol architectures.

3. Scripting & Programming languages (Python, Bash, PowerShell).
4. Malware and Antivirus analysis fundamentals.
5. Core Security concepts (CIA Triad, Least Privilege, Zero Trust).
6. Database fundamentals (SQL / NoSQL security).
7. DNS, SSL/TLS Digital Certificates, CDNs, Web Servers, and Load Balancers.

Core Assignment / Question

Question: When a user types a URL into a web browser and presses Enter, what happens behind the scenes and how does traffic flow across DNS, routers, load balancers, web servers, and databases?

5. Networking & The OSI Model

A **Computer Network** is an interconnection of computers and electronic devices enabling communication and resource sharing.

The **OSI (Open Systems Interconnection)** reference model was developed by the **ISO (International Organization for Standardization)** and defines 7 functional layers:

Layer #	Layer Name	Function & Protocols
7	Application Layer	User interface & application interaction (HTTP, HTTPS, FTP, DNS, SSH, SMTP).
6	Presentation Layer	Data formatting, translation, compression, and encryption (SSL/TLS, ASCII, JPEG).
5	Session Layer	Establishes, manages, and terminates communication sessions (RPC, NetBIOS).
4	Transport Layer	End-to-end communication, flow control, port addressing (TCP, UDP).
3	Network Layer	Logical addressing and routing across networks (IPv4, IPv6, ICMP, Routers).
2	Data Link Layer	Physical addressing, frame synchronization, error detection (MAC, Ethernet, Switches).
1	Physical Layer	Raw bit transmission over physical media (Cables, Hubs, Radio frequencies).

6. IP Addressing (IPv4) & Subnetting Basics

An **IP Address (Internet Protocol)** is a 32-bit numerical identifier assigned to each node participating in an IP-based computer network.

Classes of IPv4 Addresses

Class	First Octet Range	Default Subnet Mask	Application & Purpose
Class A	0 - 126	255.0.0.0 (/8)	Very large enterprise & government networks.
Class B	128 - 191	255.255.0.0 (/16)	Medium to large enterprise / university networks.
Class C	192 - 223	255.255.255.0 (/24)	Small business, home, and local office networks.
Class D	224 - 239	N/A	Reserved for Multicast traffic.
Class E	240 - 255	N/A	Reserved for Experimental & Research purposes.

**Note: 127.0.0.0/8 is reserved for loopback and local host diagnostics.*

Private IP vs. Public IP Comparison (RFC 1918)

Parameter	Private IP Address	Public IP Address
Network Scope	Local Area Networks (LAN) only.	Global Internet (WAN).
Internet Routing	Not routable over the public Internet.	Globally routable worldwide.
Assignment	Assigned by LAN administrator / local DHCP router.	Assigned by ISP / IANA.

Uniqueness	Unique only within the local private network.	Globally unique worldwide.
Cost	Free of charge.	Recurring cost through ISP subscriptions.
Class A Range	10.0.0.0 to 10.255.255.255	1.0.0.0 – 9.255.255.255 & 11.0.0.0 – 126.255.255.255
Class B Range	172.16.0.0 to 172.31.255.255	128.0.0.0 – 171.255.255.255 & 172.32.0.0 – 191.255.255.255
Class C Range	192.168.0.0 to 192.168.255.255	192.0.0.0 – 192.167.255.255 & 192.169.0.0 – 223.255.255.255

Subnet Mask

A **Subnet Mask** is a 32-bit number used in combination with an IP address to divide it into two portions: **Network Bits** (identifying the specific network) and **Host Bits** (identifying the specific host on that network).